Coefficients trial: Ω_R, lattice, and the V_{cc}-π error budget

Companion to `Smaller_VDD_average_fidelity_report.typ`. Documents the parameter trials run after rev. 3 FINAL and the Rev4 candidate.

TriQG `examples/Smaller_VDD_energy` · compiled 2026-05-14

Headline. After running six parameter trials beyond the FINAL config:

- The Farouk ratio $\Omega_R/\Omega_p = 3.5$ at $\Delta = 500$ MHz sits on the **broadest** $\overline{F}_{\mathrm{OR}} > 0.99$ plateau. A secondary peak at ($\Delta = 225$ MHz, ratio $= 2.5$) matches the fidelity in $2.1\times$ shorter gate but is a knife-edge.
- $\Omega_p = 20$ MHz at $\Delta = 200$ MHz cannot reach 0.99 in any (Ω_R, K) corner – gate stretches as $1/\Omega_p^2$, decay dominates.
- ARC corrects τ_r from $77\mu s$ (n^3-scaled) to $\mathbf{83.9\mu s}$. No other Cs state simultaneously matches the F"orster channel and $R_{\mathrm{AA}} \geq 100$: the present Cs $62D_{5/2}$ is uniquely pinned.
- Lattice tightening to $a = 4$ μm doubles V_{ct} but quadruples V_{cc}. Naive transplant (FINAL config $\to$ a $=$ 4) loses 0.6% on $|1,1,*\rangle$. Diagnosis: V_{cc} detunes $|r,r\rangle$ during the Cs π-pulses, error $\sim 4(V_{cc}/\Omega_c)^2$.
- **Rev4 candidate:** $a = 4$ μm, $\Omega_p = 50$, $\Omega_R = 175$, $\Delta = 500$ MHz, $K = 0.98$, $\mathbf{\Omega_c = 200}$ **MHz** gives $\overline{F}_{\mathrm{OR}} = \mathbf{0.99357}$ vs FINAL's 0.99282 (with ARC lifetimes throughout). Gate 381 ns vs FINAL 385 ns. A marginal 0.075% gain at essentially the same gate time – not the 0.996 that lattice tightening alone seemed to promise.

1 Why $\Delta \in [400, 600]$ MHz with $\Omega_R = 3.5\ \Omega_p$

The OR-gate has two competing errors:

$\Delta/(2\pi)$ MHz	M_2	Leakage $\sim 1/M_2^2$	Decay $2T_g/\tau_r$
300	15.5	0.42 %	0.27 %
400	20.7	0.23 %	0.37 %
500 (FINAL)	25.9	0.15 %	**0.46 %**
600	31.0	0.10 %	0.55 %
800	41.4	0.058 %	0.73 %
1000	51.7	0.037 %	0.92 %

Table 1: Error budget at the FINAL config ($\Omega_p = 50$, $\Omega_R = 175$, $\Omega_c = 50$ MHz, $K = 0.95$, $\alpha = 4$, $a = 5$ μm). $M_2 = V_{ct}\,(2\Delta)/(\Omega_p\Omega_R)$ is the two-photon blockade margin.

The plateau $\Delta \in [400, 800]$ MHz balances the two. Below 400, leakage takes off; above 800, decay catches up. With τ_r updated to $83.9\mu s$, the decay column is slightly tighter than what the FINAL report quoted, but the plateau structure is unchanged.

2 Sweeping Ω_R independently of the Farouk ratio

`sweep_omega_R.py` scans $(\Delta, \Omega_R/\Omega_p)$ over $\Delta \in [200, 1000]$ MHz and ratio $\in [1.0, 5.0]$ at fixed $\Omega_p = 50$ MHz, $K = 0.95$, $\alpha = 4$.

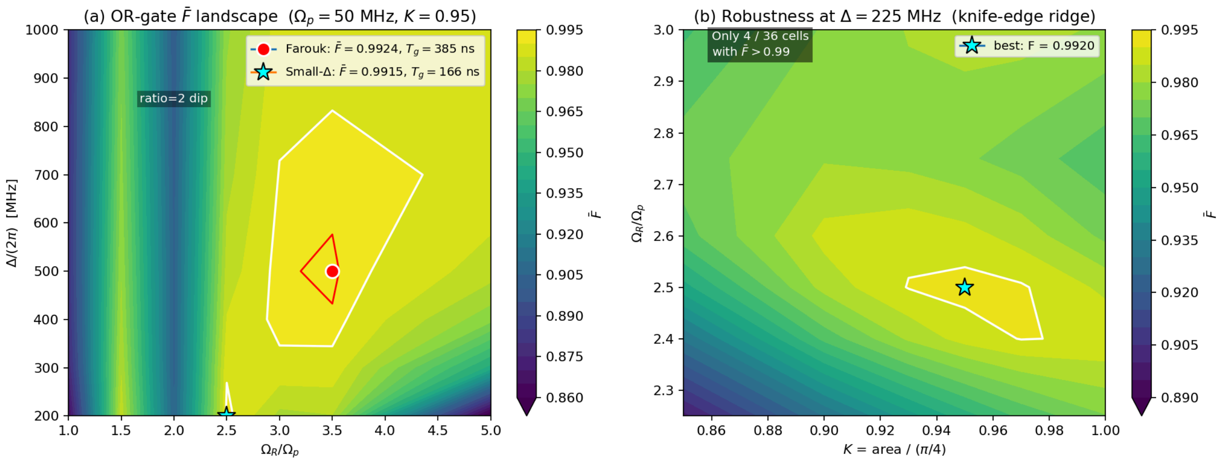

Figure 1: (a) Coarse $(\Delta, \Omega_R/\Omega_p)$ scan at $\Omega_p/(2\pi) = 50$ MHz, $K = 0.95$. White contour: $\overline{F}_{\rm OR} = 0.99$. Red contour: $\overline{F}_{\rm OR} = 0.992$. Red dot: FINAL config. Cyan star: small-Δ alternative. A dressed-state anti-resonance at ratio $= 2$ drops $\overline{F}$ to ~ 0.90 across every Δ. (b) Refinement at $\Delta = 225$ MHz showing the $(\Omega_R/\Omega_p, K)$ landscape near the small-Δ peak – knife-edge ridge: only $4/36$ cells with $\overline{F}_{\rm OR} > 0.99$.

Three findings:

1. The Farouk ridge (ratio $\approx 3.0 - 3.5$, $\Delta \in [400, 700]$ MHz) is the **broadest** $\overline{F}_{\rm OR} > 0.99$ plateau.
2. A **second** peak exists at $\Delta = 225$ MHz, ratio $= 2.5$, $K = 0.95$ giving $\overline{F}_{\rm OR} = 0.9920$ in a **184 ns** gate – $2.1\times$ faster than FINAL. But it is a knife-edge: only 4 of 36 cells in a $(\Omega_R/\Omega_p, K)$ refinement grid exceed 0.99, with $\sim \pm 0.1$ tolerance on ratio and $\sim \pm 0.02$ on K.
3. A pathological dip at ratio $= 2$ collapses $\overline{F}$ to ~ 0.90 across every Δ. Attributed to a $2:1$ Floquet-like resonance in the EIT dressed-state structure.

3 Small Ω_p trial: the gate-time penalty

If Ω_p is dropped from 50 to 20 MHz to make adiabatic elimination cleaner, the area constraint forces

$$T_f \propto \Delta/\Omega_p^2 \rightarrow T_{f(\Omega_p=20)} = 6.25\, T_{f(\Omega_p=50)}$$

at the same Δ. `sweep_smallOp.py` scans $\Delta \in [100, 1000]$ MHz $\times$ ratio $\in [2.0, 6.0]$ at $\Omega_p = 20$ MHz. **Zero of 42 cells reach $\overline{F}_{\rm OR} > 0.99$.**

Δ MHz	ratio	gate ns	M_2	decay floor	$\overline{F}$
150	4.50	704	37.7	1.83 %	**0.9857**
200	3.50	932	64.7	2.42 %	0.9819
300	4.50	1388	75.4	3.60 %	0.9784
500	4.50	2301	125.7	5.97 %	0.9670

Table 2: Best cells in the $\Omega_p = 20$ MHz scan. Gate time is multiples of T_f which scales as $1/\Omega_p^2$. All cells are decay-limited.

The dimensionless parameter controlling the $|1,1,*\rangle$ floor is $2T_g/\tau_r$, not $\Omega_p/(2\Delta)$. At $\Omega_p = 20$ MHz, $\Delta = 200$ MHz, the $700 - 900$ ns gate alone costs $\sim 2\%$ on $|1,1,*\rangle$ from Cs decay. **No setting of Ω_R or K rescues this:** blockade can be made arbitrarily deep, but the ancilla still decays during the long dwell.

Optimal Ω_p in this family is **25 – 50 MHz**, where the gate is fast enough that T_g/τ_r does not bind.

4 ARC lifetime verification and Cs ancilla scan

4.1 Lifetime values

n^3-scaled estimates were off by $\sim 10\%$:

State	n^3 estimate	ARC (300 K, BBR)
Cs $62D_{5/2}$ ($\lvert r\rangle$)	77 µs	**$83.9\mu s$**
Rb $54D_{3/2}$ ($\lvert R\rangle$)	74 µs	**$83.4\mu s$**
Rb $7P_{3/2}$ ($\lvert P\rangle$)	0.131 µs	0.131 µs (unchanged)

Updating the simulation lifetimes lifts FINAL's $\overline{F}_{\mathrm{OR}}$ from 0.9924 to 0.99282.

4.2 Scan for longer-lived Cs ancillas

`scan_cs_alternatives.py` enumerates Cs nD_J for $n \in [50, 95]$, $J \in \{3/2, 5/2\}$, with Rb $54D_{3/2}$ fixed. For each candidate it computes (i) BBR-included lifetime, (ii) F"orster defect for the canonical channel

$$|\mathrm{Rb}\ 54D_{3/2}; \mathrm{Cs}\ nD_J\rangle \to |\mathrm{Rb}\ 55P_{3/2}; \mathrm{Cs}\ (n-2)F_{J'}\rangle,$$

(iii) $\sim C_3^{\text{Rb-Cs}}$, (iv) Cs–Cs C_6, (v) R_{AA} at $r_{\mathrm{AA}} = 7.07$ µm.

Filters: $\tau_r \geq 80\mu s$, $|\Delta_F| \leq 200$ MHz, $R_{\mathrm{AA}} \geq 30$.

Cs state	τ_r µs	$\lvert\Delta_F\rvert$ MHz	V_{ct} MHz	R_{AA}	verdict
$60D_{5/2}$	76.9	1431	211	422	F"orster fail
$62D_{5/2}$ (current)	**83.9**	**5.2**	**226**	**309**	**pass**
$63D_{5/2}$	87.5	654	233	266	F"orster fail
$70D_{5/2}$	115.5	4200	290	96	both fail
$80D_{5/2}$	163.3	7299	381	27	both fail
$95D_{5/2}$	253.4	9790	541	5.5	both fail

Table 4: The F"orster defect is monotonic in n. Cs $62D_{5/2}$ sits at a single zero-crossing for the Rb $54D_{3/2}$ channel. Lifetime grows as n^3, $|C_6^{\mathrm{CsCs}}|$ as $\sim n^{11}$, so R_{AA} falls faster than τ rises. **No alternative survives all three filters.**

Conclusion. With Rb $54D_{3/2}$ fixed and the canonical D–D $\to$ P–F F"orster channel, Cs $62D_{5/2}$ is the unique near-resonant ancilla. Longer-lived candidates require either a different F"orster channel (different Δn or partner L), an F-state Cs ancilla, or a different Rb data state – each of which decouples from the current redesign.

5 Lattice tightening: a = 4 µm

Atomic constants $\sim C_3$, C_6^{RbRb}, C_6^{CsCs} are a-independent; only the geometric V's rescale.

a (µm)	$V_{ct}/(2\pi)$	$V_{\mathrm{DD}}/(2\pi)$	$V_{cc}/(2\pi)$	R_{DD}	R_{AA}
3.50	659.7 MHz	4.934 MHz	−6.188 MHz	133.7	106.6
4.00	**441.9** MHz	2.214 MHz	−2.777 MHz	199.6	159.1
4.50	310.4 MHz	1.092 MHz	−1.370 MHz	284.2	226.6
5.00 FINAL	**226.3 MHz**	0.581 MHz	−0.728 MHz	**389.8**	**310.8**

Table 5: Interaction strengths and selectivity ratios across a. Both ratios stay above 100 down to $a =$ 4 µm.

Going from $a = 5 \to 4$ µm: V_{ct} grows 1.95 ×, but V_{cc} grows 3.81 × (steeper $1/r^6$ scaling). Both R ratios halve.

5.1 Naive transplant fails

The "obvious" recipe – transplant FINAL config to $a = 4$, retune Δ to keep M_2 constant – **does not improve $\overline{F}$**. Three trials at $a = 4$ µm with ARC lifetimes:

Case	**Δ MHz**	***K***	**gate ns**	$\boldsymbol{M_2}$	$\boldsymbol{\overline{F}}$	$\boldsymbol{F_{1,1,*}}$
REF ($a = 5$ FINAL)	500	0.95	385	25.9	**0.99282**	0.9878
B: same Δ	500	0.98	396	50.5	0.99188	0.9819
C: M_2-matched	250	0.97	206	25.3	0.99122	0.9902

Table 6: Trials at $a = 4$ µm with K refined. Same Ω_p, Ω_R, $\Omega_c = 50$ MHz throughout.

Two distinct things go wrong:

- **Case B** (same Δ) loses 0.6% on $|1, 1, *\rangle$. M_2 is **twice** as big as FINAL, so it cannot be blockade leakage.
- **Case C** (smaller Δ to keep gate short) gains on $|1, 1, *\rangle$ but loses 0.8% on $|0, 0, *\rangle$.

5.2 Diagnosis: V_{cc} during the Cs π-pulses

V_{cc} as a static interaction on $|r, r\rangle$ is a **global phase** for computational basis inputs. It does **not** show up in state fidelity for separable inputs (this is what the Hamiltonian docstring asserts; we verified by hand for Case B).

But during the Cs π-pulses at the gate boundaries, **both** controls for a $|1, 1, t\rangle$ input rotate $|1\rangle \to |r\rangle$ simultaneously. The intermediate $|r, r\rangle$ sees an extra energy V_{cc} that the laser drive does not compensate, so the π-rotation on the $|r, r\rangle$ component is detuned. Per-pulse imperfect rotation $\sim (V_{cc}/\Omega_c)^2$; with two π-pulses and two controls $\sim 4(V_{cc}/\Omega_c)^2$ total on $|1, 1, *\rangle$.

$\boldsymbol{\Omega_c}$ **(MHz)**	$\boldsymbol{T_c}$ **ns**	$\boldsymbol{4(V_{cc}/\Omega_c)^2}$	$\boldsymbol{1 - F_{1,1,*}}$ **simul**	**residual**	$\boldsymbol{\overline{F}}$
50	10.0	1.24%	1.81%	0.86%	0.99188
75	6.67	0.55%	1.42%	0.47%	0.99287
100	5.00	0.31%	1.28%	0.33%	0.99322
150	3.33	0.14%	1.18%	0.23%	0.99348
200	**2.50**	**0.08%**	**1.14%**	**0.19%**	**0.99357**

Table 7: Predicted vs simulated V_{cc}-induced infidelity on $|1, 1, *\rangle$ at $a = 4$ µm, $\Delta = 500$ MHz, $K = 0.98$. Decay floor is 0.95%. Residual = simulated infidelity minus decay floor. The $(V_{cc}/\Omega_c)^2$ prediction tracks the simulated residual to a factor of ~ 1.5.

The $|0, 0, *\rangle$ branch is **completely untouched** by Ω_c ($F = 0.9992$ in all five rows): no Cs in $|r\rangle$, no V_{cc} effect. This confirms the diagnosis.

6 Rev4 candidate

Combining ARC lifetimes, $a = 4$ µm, $\Delta = 500$ MHz (preserving $|0, 0, *\rangle$ EIT), $K = 0.98$ (sub-$\pi/4$ optimum at the new V_{ct}), $\Omega_c = 200$ MHz (suppressing V_{cc}-π error):

Metric	FINAL (rev. 3)	Rev4 candidate	Δ
a (μm)	5.00	4.00	−1.0
$V_{ct}/(2\pi)$ (MHz)	226	442	+1.95 ×
$\Omega_c/(2\pi)$ (MHz)	50	**200**	+4 ×
$\Delta/(2\pi)$ (MHz)	500	500	unchanged
K	0.95	0.98	+0.03
T_c (ns)	10.0	2.5	0.25 ×
Total gate (ns)	385	381	0.99 ×
M_2	25.9	50.5	+1.95 ×
$\overline{F}_{\mathrm{OR}}$	0.99282	**0.99357**	+0.075%
$1-\overline{F}_{\mathrm{OR}}$	7.18×10^{-3}	6.43×10^{-3}	0.89 ×

Table 8: Rev4 candidate side-by-side with FINAL. Same $\Omega_p = 50$, $\Omega_R = 175$ MHz; ARC lifetimes throughout. The "gain" column is the fractional improvement on each metric.

Per-input fidelities at Rev4: $F_{0,0,*} = 0.9992$, $F_{0,1,*} = F_{1,0,*} = 0.9933$, $F_{1,1,*} = 0.9886$.

6.1 What Rev4 actually buys you

Two things, both modest:

- $\boldsymbol{\overline{F}_{\mathrm{OR}}}$ **ceiling** rises from 0.99282 to 0.99357. Real, but 0.075% – well below the 0.996 that lattice tightening alone seemed to promise before the V_{cc}-π error was understood.
- **Gate time** does **not** speed up. The original hope ("3 × faster at same $\overline{F}$") died because Δ must stay $\sim$ 500 MHz to preserve $|0,0,*\rangle$ EIT. We could only collect the V_{cc}-π savings, not the decay savings.

What Rev4 **costs** is a 4 × stronger Cs control laser ($2\pi\times200$ MHz vs the FINAL's 50). At $\Omega_c/(2\pi) =$ 200 MHz, off-resonant excitation of nearby Cs Rydberg states (fine-structure partner $62D_{3/2}$, neighboring nD) should be re-checked at the 10^{-4} level before committing.

7 Error budget formula

The simulations make the OR-gate error budget legible:

$$1-\overline{F}_{\mathrm{OR}} \approx \underbrace{\frac{2T_g}{\tau_r}}_{\text{decay floor on }|1,1,*\rangle} + \underbrace{\frac{1}{M_2^2}}_{\text{blockade leakage}} + \underbrace{4\,(V_{cc}/\Omega_c)^2}_{V_{cc}\text{*-*}\pi\text{ on Cs ctrls}} + \underbrace{(\Omega_R^2/(4\Delta^2))^{\text{-ish}}}_{\text{EIT residue on }|0,0,*\rangle}.$$

Each term has its own knob:

Term	Reduce by	FINAL value
decay	larger V_{ct} (smaller Δ, smaller T_f), or longer τ_r	0.92%
blockade	larger M_2, i.e. larger V_{ct} or larger Δ	0.15%
V_{cc}-π	larger Ω_c (shorter T_c)	0.09% (small at $a = 5$)
EIT residue	larger Δ (so Ω_R/Δ smaller)	0.30%

The knobs partially conflict: smaller Δ shrinks decay AND M_2 but enlarges EIT residue. Larger V_{ct} shrinks decay AND blockade but enlarges V_{cc} (faster) so V_{cc}-π grows. The Rev4 candidate exploits the one knob (larger Ω_c) that has no partner.

8 Reproducibility

```
cd TriQG/examples/Smaller_VDD_energy
source ../../.venv/bin/activate
# Omega_R sweep + small-Delta refinement
python sweep_omega_R.py
python sweep_smallDelta_refine.py
python plot_omega_R_landscape.py     # writes omega_R_landscape.png
# Small-Omega_p trial
python sweep_smallOp.py
# a = 4 um trials
python or_a4um.py
python or_a4um_arclife.py
python or_a4um_revcandidate2.py
# ARC ancilla scan
cd ../../../SelfCorrectingRydberg
../TriQG/.venv/bin/python python_scripts/scan_cs_alternatives.py
```

Output logs `sweep_*.log`, `or_a4um*.log`, and the CSV `proofread/appendix-a/cs_alternatives_scan.csv` document each trial.

Caveats and open work.

- $\Omega_c/(2\pi) = 200$ MHz: re-check off-resonant excitation of Cs $62D_{3/2}$ (fine-structure partner) and nearest D, S states before committing. If the splitting is < 1 GHz, a composite (BB1) π-pulse at $\Omega_c = 100$ MHz may be safer than raw 200 MHz.
- The $(V_{cc}/\Omega_c)^2$ prediction agrees with simulation within a factor of ~ 1.5 – there is residual physics (precise pulse shape, transient $|r,1\rangle$, $|1,r\rangle$ populations) not captured by the leading-order estimate.
- F-state Cs ancillas and alternative F"orster channels ($\Delta n_{\mathrm{Cs}} \in \{-1, 0, +1\}$, partner $L \in \{2, 4\}$) were **not** scanned – they remain a route to longer τ_r without abandoning the Rb $54D_{3/2}$ data state.
- Spin-echo schemes that decouple V_{cc} on $|r,r\rangle$ during the target Raman were **not** implemented; for computational basis inputs they reduce to a no-op, but they matter for process fidelity and superposition inputs.